

EN3160 Tutorial 2

Ranga Rodrigo

August 11, 2026

1. Total least squares is useful in vision problems.

- (a) Show how to find the line parameters expressed as $ax + by = d$ using this method in terms of point coordinates (x_i, y_i) , $i = 1, \dots, N$, and U , in regular notation.

$$\text{Loss: } E = \sum_{i=1}^N |ax_i + by_i - d|^2$$

$$\frac{\partial E}{\partial d} = -2 \sum_{i=1}^N [ax_i + by_i - d] = 0$$

$$a \sum_{i=1}^N x_i + b \sum_{i=1}^N y_i - Nd = 0$$

$$d = a\bar{x} + b\bar{y}$$

$$\|n\|^2 = \left\| \begin{pmatrix} a \\ b \end{pmatrix} \right\|^2 = a^2 + b^2 = 1$$

let a & b be the smallest eigenvectors of smallest

$$E = \sum_{i=1}^N |a(x_i - \bar{x}) + b(y_i - \bar{y})|^2 \quad \text{eigen value of } E$$

$$= (U_n)^T (U_n)$$

$$= n^T U^T U n$$

$$U^T U n = \lambda n \quad (\text{Using Lagrange multipliers})$$

$$E = n^T \lambda n = \lambda n^T n = \lambda$$

$$U = \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ x_2 - \bar{x} & y_2 - \bar{y} \\ \vdots & \vdots \\ x_N - \bar{x} & y_N - \bar{y} \end{bmatrix}, \quad n = \begin{bmatrix} a \\ b \end{bmatrix}$$

- (b) In a particular case, $\bar{x} = 5.0$, $\bar{y} = 13.1$. The eigenvalues of $U^T U$ are 21.2, 4324.1 and corresponding eigenvectors are $[-0.8974, 0.4411]^T$, and $[-0.4411, -0.8974]^T$. Find the gradient m and intercept c of the total-least-squares-fit line.

$$a = -0.8974$$

$$b = 0.4411$$

$$d = a\bar{x} + b\bar{y}$$

$$= (-0.8974)(5) + (0.4411)(13.1)$$

$$= 1.29141$$

$$ax + by = d$$

$$y = -\frac{a}{b}x + \frac{d}{b}$$

$$c = \frac{d}{b} = \frac{(1.29141)}{(0.4411)} = 2.9277$$

$$m = -\frac{a}{b} = 2.034$$

2. Figure 1 shows the image 1 and 4 from the graffiti images, and the homography that maps image 1 to 4. Note the convention of axes.

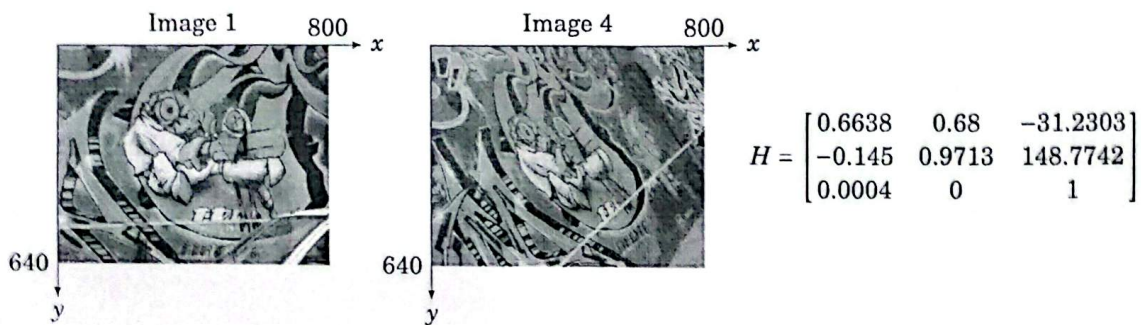


Figure 1: Image 1, image 4 and homography mapping image 1 to 4.

- (a) State the transformation that maps image 4 to image 1.

$$\begin{aligned} X_2 &= HX_1 \quad \therefore \text{The transformation:} \begin{bmatrix} 1.2203 & -0.8543 & 165.2121 \\ 0.2569 & 0.8496 & -118.3844 \\ -0.0005 & -0.00034 & 0.9339 \end{bmatrix} \\ X_1 &= H^{-1}X_2 \quad \text{is } H^{-1} \end{aligned}$$

- (b) Compute the locations of the corners of image 4 when mapped onto image 1.

$$\begin{aligned} \text{Image 2} \Rightarrow \quad TL &= H^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 165.2121 \\ -118.3844 \\ 0.9339 \end{bmatrix} = \begin{bmatrix} 165.2121 & -118.3844 \\ 0.9339 & 0.9339 \end{bmatrix} \Rightarrow (176.9, -126.8) \\ \text{Image 1} \Rightarrow TL &= (0, 0) \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ TR &= (800, 0) \Rightarrow \begin{bmatrix} 800 \\ 0 \\ 1 \end{bmatrix} \\ BL &= (0, 640) \Rightarrow \begin{bmatrix} 0 \\ 640 \\ 1 \end{bmatrix} \\ BR &= (800, 640) \Rightarrow \begin{bmatrix} 800 \\ 640 \\ 1 \end{bmatrix} \\ TR &= H^{-1} \begin{bmatrix} 800 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1141.4554 \\ 87.1661 \\ 0.5434 \end{bmatrix} \Rightarrow (2100.4, 160.4) \\ BL &= H^{-1} \begin{bmatrix} 0 \\ 640 \\ 1 \end{bmatrix} = \begin{bmatrix} -381.5564 \\ 425.4028 \\ 1.1526 \end{bmatrix} \Rightarrow (-331, 369.1) \\ BR &= H^{-1} \begin{bmatrix} 800 \\ 640 \\ 1 \end{bmatrix} = \begin{bmatrix} 594.65 \\ 630.95 \\ 0.7621 \end{bmatrix} \Rightarrow (786.3, 827.9) \end{aligned}$$

- (c) What is the size of the arrays needed to accommodate the stitched image after this mapping?

$$\begin{aligned} \text{Image 1: } x \text{ coordinates} &= [0, 800], y \text{ coordinates} = [0, 640] \\ \text{Image 4: } x \text{ coordinates} &= [-331, 2100.4], y \text{ coordinates} = [-126.8, 827.9] \\ \therefore \text{Array Array size} &\begin{cases} \text{Width: } 2100.4 - (-331) = 2431.4 \text{ pixels} \\ \text{Height: } 827.9 - (-126.8) = 954.7 \text{ pixels} \end{cases} \end{aligned}$$

- (d) What are two additional steps to make the stitched result visually pleasing?

- 1) Image Blending
 $\Rightarrow$ This eliminates the visible seam in the overlapping region
- 2) Exposure compensation
 $\Rightarrow$ By adjusting exposure or colour correcting for the two images.

Number:

2 3 0 5 6 6 0

3. The scale-normalized Laplacian of Gaussian (LoG) for circularly symmetric blob detection is

$$V_{\text{norm}}^2 g = \sigma^2 \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \right),$$

where g is the bi-variate Gaussian.

- (a) Obtain an expression for the radius of a black circle in a white background that would give the maximum response to this operator.

$$\begin{aligned} \frac{\partial g}{\partial x} &= -\frac{x}{\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}} & \frac{\partial g}{\partial y} &= 0 \\ \frac{\partial^2 g}{\partial x^2} &= \frac{x^2 - \sigma^2}{\sigma^4} g(x) & \Rightarrow & \left(\frac{x^2 + y^2 - 2\sigma^2}{\sigma^4} g(x) \right) \neq 0 \\ \frac{\partial^2 g}{\partial y^2} &= \frac{y^2 - \sigma^2}{\sigma^4} g(x) & \Rightarrow & x^2 + y^2 = 2\sigma^2 \\ & & & \Rightarrow r = \sqrt{2}\sigma \end{aligned}$$

- (b) What is the suitable value for σ , if black circles of radius 10 pixels are to be detected.

$$\begin{aligned} \sqrt{2}\sigma &= 10 \\ \sigma &= 7.0711 \end{aligned}$$

- (c) Assume that two images of the scene—taken with two level of zoom—have features detected. How can the features be matched?

After knowing the characteristic scale (σ) of blobs in each image, both images can be normalized to a standard reference size. Now features can be matched.

4. A conveyor belt carries several types wrapped chocolate bars. Images of reference wrapped chocolate bars are available. The factory has the following requirements:

- Counting the number of wrapped chocolate bars that pass along the belt.
- Determining if there are printing mistakes in comparison with references.

- (a) Propose a system for counting.

Mount a camera directly above the conveyor belt. Apply Gaussian Blur to remove noise and take a video stream. Since the camera & belt are not moving relative to each other, select ROI such that belt is missing in every frame. Apply a binary mask to separate the bar from surrounding. Run canny edge detector on it. Find centroid using the edges. Draw a horizontal line splitting the ROI in the middle. Count each centroid moving pass the horizontal line.

- (b) Propose a method for locating printing mistakes.

Capture an image of the target and detect features like blobs or contours. Compare these features with the reference images. Use the matched features to align the captured images to reference geometrically. Subtract the aligned image from reference to get pixel intensity mismatches. Use a threshold to remove noise and a binary mask using a binary mask. Identify printing mistakes by locating clusters of errors in the binary mask.

5. Consider the set of points that corresponds to a part of a corner of a box

$$X = \begin{bmatrix} 0 & 1.277 \\ 0 & -0.796 \\ 3 & -2.016 \\ 3 & 2.196 \\ 6 & -2.907 \\ 6 & 6.594 \\ 9 & -7.897 \\ 9 & 6.373 \\ 12 & -10.023 \\ 12 & 12.723 \end{bmatrix}$$

also shown in shown in Fig. 2. The figure includes a least-squares fitted line and thresholds used to determine the consensus.

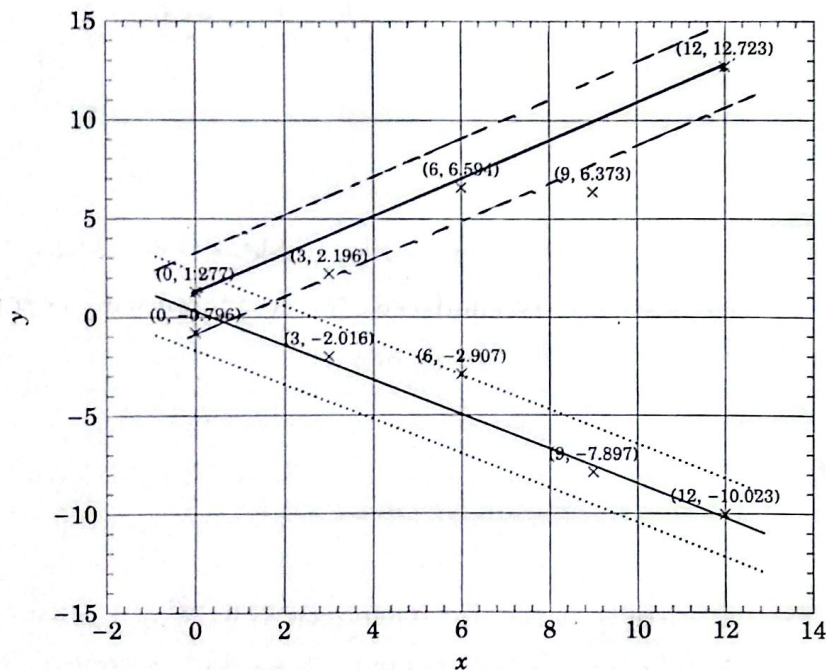


Figure 2: Points for line fitting.

(a) State how a second line can be fitted.

By removing the points that formed the consensus set for the first fitted line and then applying RANSAC to the rest of the points to find the second line.

(b) Using the RANSAC method, find an expression for this line. You may hand-pick the sample by observation. Run only one iteration. Use sketches to illustrate.

Sample: $[(0, 1.277), (12, 12.723)]$

$$m = \frac{12.723 - 1.277}{12 - 0} = \frac{11.446}{12} \therefore y = 0.954x + 1.277$$

$$= 0.954$$

$$c = 1.277$$

- (c) Briefly describe the process of fitting a circle and lines to a (different) set of points.

For a circle, unlike a line we need 3 points.

Therefore, randomly pick 3 points and calculate the circle's center and radius. Find the consensus set for that circle by counting points that are within the threshold distance from the edge of the circle. Do this until the circle with maximum inliers are found. Remove those inliers and apply RANSAC again picking 3 points randomly to find the lines.

- (d) The third edge of the box is vertical and has generated a set of noisy points. How would you fit a line to these points?

Choose a random point and draw a vertical line from that point.

Count the inliers inside the threshold region. Repeat this.

Until the line with the most consensus set is obtained.

6. Consider the scale space shown in Fig. 3.

- (a) Identify the scale-space extrema by observation, and mark them in the figure.

Marked in the figure.

- (b) Compute the radii of the blobs that correspond to each of these extrema.

$$\epsilon = 2 \Rightarrow r = 2\sqrt{2} = 2.83 \text{ pixels}$$

$$\epsilon = 3 \Rightarrow r = 3\sqrt{2} = 4.24 \text{ pixels}$$

- (c) What must be done to be able to match images of the same object taken from different viewpoints?

Find the scale space Extrema and their corresponding ϵ values for the object in each image. Crop the object by selecting the area around the extremas and normalize them using ϵ values.

Apply rotational normalization as well for each area by aligning them to their dominant gradient direction. Extract features from the normalized patches and compare them to find matching points.

7. In computing the homography using point matches between two images through RANSAC, there were 1600 inliers.

- (a) What is the minimum number of point matches required to compute the homography?

Four

- (b) Show the format of the "data matrix" used for homography computation using singular value decomposition (SVD).

Data Matrix =

$$\begin{bmatrix} -x_1 & -y_1 & -1 & 0 & 0 & 0 & x'_1 x_1 & x'_1 y_1 & x'_1 \\ 0 & 0 & 0 & -x_1 & -y_1 & -1 & y'_1 x_1 & y'_1 y_1 & y'_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -x_N & -y_N & -1 & 0 & 0 & 0 & x'_N x_N & x'_N y_N & x'_N \\ 0 & 0 & 0 & -x_N & -y_N & -1 & y'_N x_N & y'_N y_N & y'_N \end{bmatrix}$$

where (x'_i, y'_i) in image 1 maps to (x_i, y_i) in image 2

$N = 1600$

116	110	99	86	75	66	62	65	74	80	82	76	61	40	23	12	-8	-6	-6	-6	-6
110	101	81	61	46	-36	33	-36	-44	-54	-63	-66	-58	-41	-24	-13	-8	-6	-6	-6	-6
99	81	42	-4	18	27	28	27	21	5	-20	-44	-53	-44	-28	-15	-9	-6	-6	-6	-6
86	61	4	47	71	75	73	75	74	59	23	-20	-46	-47	-33	-18	-10	-7	-6	-6	-6
75	-46	18	71	89	86	81	86	94	86	49	-3	-41	-49	-37	-21	-11	-7	-6	-6	-6
66	-36	27	75	86	76	68	76	91	91	59	7	-36	-49	-40	-24	-13	-8	-7	-7	-7
62	-33	28	73	81	68	60	69	87	90	61	9	-36	-52	-43	-28	-17	-12	-11	-10	-9
65	-36	27	75	86	76	69	77	92	91	58	4	-41	-58	-51	-37	-26	-21	-19	-18	-17
74	44	21	74	91	87	92	98	88	48	-9	-53	-68	-60	-47	-38	-34	-33	-33	-32	-32
80	-54	5	59	86	91	90	91	88	68	24	-29	-66	-75	-64	-49	-40	-40	-45	-50	-52
82	63	20	23	49	59	62	59	50	26	-13	-53	-75	-71	-50	-27	-18	-24	-43	-61	-69
76	66	-44	-20	-3	7	10	7	4	-23	-49	-70	-73	-49	-11	23	35	20	-16	-53	-69
61	-58	-53	-46	-41	-36	34	36	-44	-55	-68	-73	-57	-14	42	87	102	79	27	-29	-53
-40	-41	44	-47	-49	-49	-49	-50	-54	-59	-65	-62	-35	20	87	138	154	127	65	-1	-31
-23	-24	-28	-33	-37	-39	-39	-40	-42	-47	-53	-49	-22	35	105	156	173	145	80	10	-21
-12	-13	-15	-18	-21	-23	-23	-24	-26	-32	-41	-42	-22	28	90	139	154	127	65	-1	-31
-8	8	9	-10	-11	-12	-12	-12	-15	-21	-32	-39	-30	3	50	90	102	79	27	-29	-53
-6	-6	6	-7	-7	-7	-7	-7	-10	-15	-25	-36	-38	-24	2	28	36	20	-17	-54	-70
-6	-6	-6	-6	-6	-6	-6	-6	-8	-12	-20	-30	-39	-41	-34	-24	-20	-29	-47	-64	-72
-6	-6	-6	-6	-6	-6	-6	-6	-7	-9	-15	-24	-35	-44	-49	-50	-51	-53	-58	-62	-63
-6	-6	-6	-6	-6	-6	-6	-6	-7	-9	-13	-21	-32	-44	-53	-58	-59	-60	-60	-59	-58

Scale
Space
Extrema

(b) $\sigma = 2$.

-97	-91	-73	-48	-21	2	16	19	12	-2	-21	-40	-55	-65	-67	-63	-55	-45	-37	-32	-30
-91	-84	-66	-41	-14	9	23	26	19	3	-16	-36	-53	-63	-66	-63	-55	-46	-38	-33	-31
-73	-66	-48	-22	5	28	42	44	36	18	-4	-26	-46	-59	-64	-63	-57	-49	-42	-37	-35
-48	-41	-22	5	33	55	69	70	60	40	14	-12	-36	-53	-62	-63	-59	-53	-47	-42	-41
-21	-14	4	5	33	61	84	96	97	84	61	32	2	-26	-47	-60	-64	-62	-57	-52	-48
2	9	28	56	84	107	119	118	103	78	46	13	-18	-42	-57	-64	-64	-61	-57	-54	-53
16	23	42	69	97	119	130	128	113	86	53	19	-13	-38	-55	-62	-64	-62	-59	-57	-56
20	26	45	71	98	119	129	126	111	85	52	19	-12	-35	-50	-58	-59	-58	-56	-55	-54
13	19	37	61	86	105	114	111	96	73	43	14	-12	-31	-43	-48	-49	-48	-47	-46	-46
-1	4	20	42	64	81	88	86	73	53	29	6	-13	-25	-32	-33	-32	-31	-30	-31	-31
-20	-15	-2	17	36	50	56	54	44	29	12	-2	-12	-17	-16	-12	-9	-7	-7	-8	-9
-38	-34	-24	-9	16	17	22	21	14	5	-4	-10	-10	-6	2	11	17	20	19	18	17
-53	-50	-42	-31	-20	-12	-9	-10	-13	-17	-19	-16	-8	19	19	33	41	44	43	41	40
-62	-60	-55	-47	-40	-35	-33	-35	-35	-31	-21	6	13	32	48	58	61	60	57	57	55
-64	-63	-60	-55	-51	-49	-49	-49	-46	-38	-25	-6	16	37	54	64	66	64	60	59	59
-60	-59	-58	-55	-54	-54	-55	-56	-56	-52	-43	-29	10	12	32	48	57	58	55	51	49
-52	-51	-51	-50	-51	-52	-55	-57	-57	-54	-46	-34	-17	2	19	32	38	38	34	29	27
-42	-42	-42	-43	-45	-47	-51	-54	-55	-54	-49	-39	-26	-12	1	10	14	12	7	2	0
-34	-34	-35	-36	-39	-42	-46	-50	-53	-53	-51	-45	-36	-26	-17	11	-11	-14	-14	-20	-25
-28	-28	-29	-31	-34	-38	-42	-47	-51	-53	-52	-49	-43	-36	-30	-27	-28	-33	-39	-45	-47
-26	-27	-28	-29	-32	-36	-41	-46	-50	-52	-53	-50	-46	-40	-35	-33	-35	-40	-46	-52	-54

Scale —
Space
Extrema

(d) $\sigma = 4$.

(c) When SVD is used the resulting V^T is

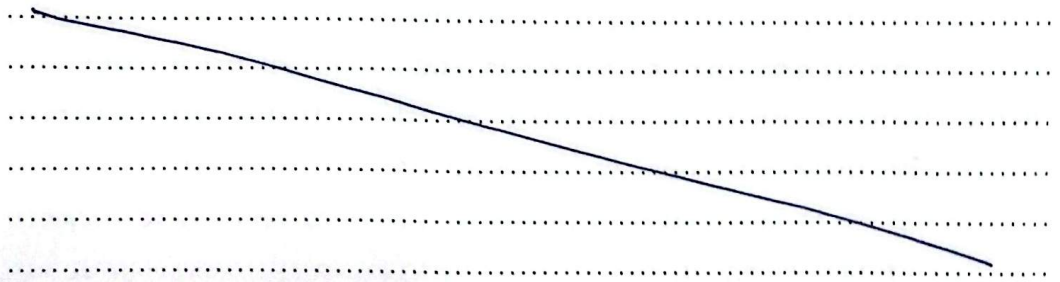
$$\begin{bmatrix} 1.0 \times 10^{-3} & 4.2 \times 10^{-4} & 1.4 \times 10^{-6} & 7.0 \times 10^{-4} & 3.9 \times 10^{-4} & 1.1 \times 10^{-6} & -9.0 \times 10^{-1} & -4.3 \times 10^{-1} & -1.3 \times 10^{-3} \\ 1.3 \times 10^{-3} & -5.0 \times 10^{-4} & 1.1 \times 10^{-6} & -4.6 \times 10^{-4} & -1.3 \times 10^{-3} & -2.1 \times 10^{-6} & -4.3 \times 10^{-1} & 9.0 \times 10^{-1} & 4.6 \times 10^{-4} \\ 5.1 \times 10^{-1} & 2.8 \times 10^{-1} & 8.4 \times 10^{-4} & -7.6 \times 10^{-1} & -3.0 \times 10^{-1} & -1.1 \times 10^{-3} & 5.4 \times 10^{-4} & -1.2 \times 10^{-3} & -2.2 \times 10^{-3} \end{bmatrix}$$

Optimal vector is always found in the last row of V^T according to SVD

$$\therefore H = \begin{bmatrix} 0.51 & 0.28 & 8.4 \times 10^{-4} \\ -0.76 & -0.3 & -1.1 \times 10^{-3} \\ 5.4 \times 10^{-4} & -1.2 \times 10^{-3} & -2.2 \times 10^{-3} \end{bmatrix}$$

Number:

2	3	0	5	6	6	U
---	---	---	---	---	---	---



8. Fig. 4 shows the results of blob detection as scale-space extrema. The circles representing blobs along with their radii are separately shown for clarity.

Detected Extrema in Scale-Space

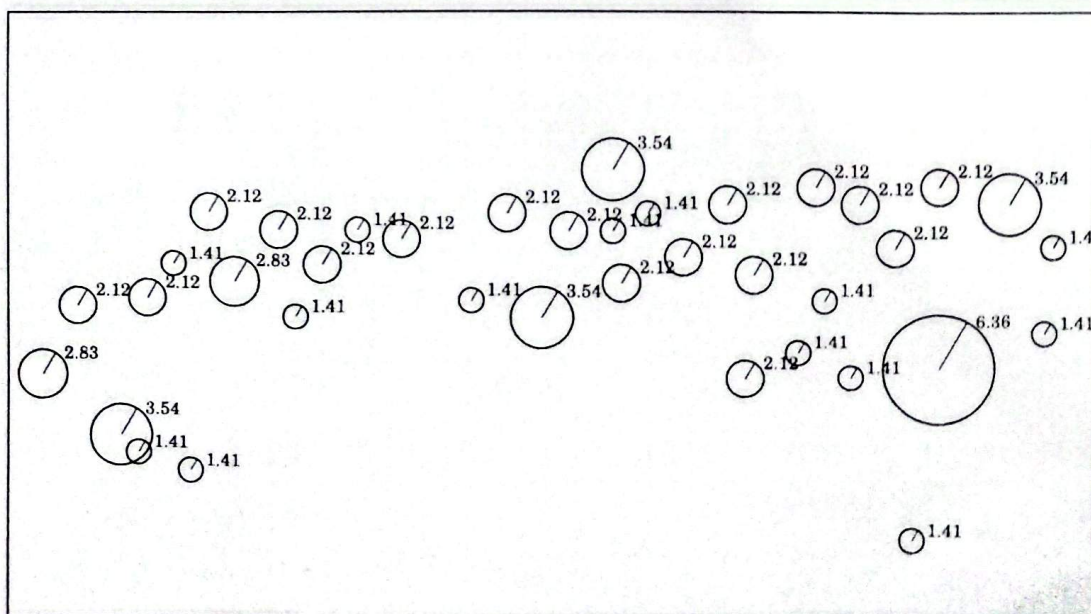
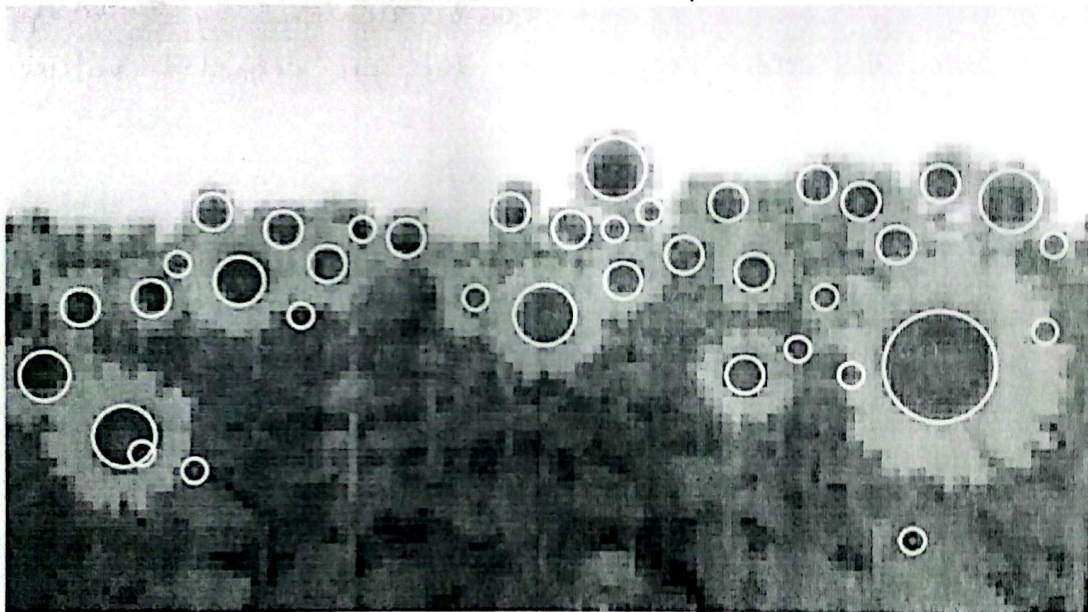


Figure 4: Blob detection.

(a) What is the reason for circles having a few specific radii?

The scale space $\mathcal{B}$ built using a discrete set of images. These images have a pre-defined blur levels (σ values). Because the radius is tied to the σ value by $r = \sqrt{2} \sigma$, radii are also have a discrete set of values possible to them.

(b) Some circles fail to fit well to the central back disk of the flower. How can this be corrected?

By using sub-scale interpolation to estimate the best fitting blob radius. This can be achieved by Taylor approximation using the filter responses for the pre-defined σ values.

(c) In scale-space extrema detection, a large number of extrema are detected. Suggest a remedy for this.

Contrast thresholding.

→ Remove any extrema that have the peak of the response of the filter, too weak to avoid noises in the image.